

The plateau versus $2\pi e$

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Abstract

The published plateau is $\Delta_\infty \approx 16.0 \pm 1.6$ on $\ell \in [40, 80)$ and 16.0 ± 2.1 on $[80, 150)$. The constant $2\pi e \approx 17.079$ sits inside both bands. New χ_3 ladders at $\mu = 16, 22, 30$ climb $15.88 \rightarrow 16.92 \rightarrow 17.07$ at levels 40, 62, 105. The deepest new spacing matches $2\pi e$ to 0.01 nat. The lock “the plateau stays at 16” is dead. $2\pi e$ survives; one character and one basis do not select it.

1 The lock

A1 said the leftover Rayleigh problem drops by a universal factor once the desert is spent. The published factor is e^{-16} . Among the small list of constants that could have been that factor, $2\pi e \approx 17.079$ is the only one inside the plateau bands and not already killed ($s = \gamma_1^2/(2\pi e)$ died on χ_4).

2 New rungs

`code/scan_s.py` on χ_3 (odd, $s = 4$, depth-adequate at these windows). Spacings $\ell_k - \ell_{k+1}$ at the deep end:

μ	N	deepest (ℓ, Δ)	$2\pi e - \Delta$
16	37	(39.5, 15.88)	+1.20
22	41	(62.3, 16.92)	+0.16
30	45	(104.7, 17.07)	+0.01

The sequence climbs toward $2\pi e$, not toward 16. At $\mu = 30$ the match is 17.07 against 17.079.

The rest of each ladder still follows the published ramp (9–13 nats on $\ell < 40$). Only the deepest spacing is the plateau test.

3 Second character, next window

`scan_s.py` omitted the primes 37+ — the accidental quorum of notebook 16.4. With the sieve extended and eigenvalues sorted, at $N = 53$:

χ	μ	deepest (ℓ, Δ)	$2\pi e - \Delta$
χ_3	38	(128.8, 16.82)	+0.26
χ_4	30	(81.4, 17.65)	−0.57
χ_4	38	(103.7, 18.16)	−1.08

χ_3 hovers 16.8–17.1 across four windows. χ_4 sits 0.6–1.1 nats above $2\pi e$. The plateau is the interval $[16.8, 18.2]$, not a selected constant. $2\pi e$ remains inside the interval; it is no longer the unique occupant. The $N = 75$ lock is still the next stroke, not a requirement to kill 16.